Course Code Course Title L T P J C

**ECE1005 Digital Logic Design Laboratory 0 0 2 0 1**

**Pre-requisite** NIL

## Course Objectives:

1. To verify the truth tables of logic gates.
2. To design and test combinational circuits.
3. To design sequential circuits using flip-flops.

## Course Outcomes:

1. Verify the operation of logic gates and Boolean identities.
2. Design adders, multiplexers and decoders.
3. Build counters and registers using flip-flops.

## List of Experiments

1. Verification of logic gates and universal gates.
2. Realization of Boolean expressions using NAND gates only.
3. Design of half adder and full adder.
4. Design of 4-bit parallel adder and subtractor.
5. Multiplexers and demultiplexers.
6. Encoders and decoders, seven segment display.
7. Flip-flops: SR, JK, D and T.
8. Design of synchronous and asynchronous counters.
9. Shift registers and ring counters.
10. Design of a simple traffic light controller.

**Total Laboratory Hours** 30 hours

Recommended by Board of Studies 05-06-2021